

# Fathan Yazid Satriani

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🌐 [LinkedIn](#) | 🐙 [GitHub](#) | 📁 [Portfolio](#)

## EDUCATION

### Universitas Indonesia

Bachelor of Engineering in Computer Engineering; GPA: 3.75/4.00

Depok, Indonesia

Exp. Apr 2027

### SMAN 34 Jakarta

Diploma of Mathematics and Sciences; Grade: 90/100

South Jakarta, Indonesia

July 2020 – Apr 2023

## EXPERIENCE

### National Taiwan University of Science and Technology (NTUST)

Research Intern, BMW Lab (TEEP Program)

Taipei, Taiwan

Jun 2026 – Aug 2026

- Completed six months of remote research (Jan–Jun), beginning with OSC SMO–FlexRIC integration over O1/A1/E2 interfaces and progressing to OAI/NVIDIA Aerial PRACH security analysis, simulation, and RFSim baselines.
- Reproduced and analyzed PRACH attacker behavior across five authorized Open RAN experiment paths using OAI, OCUDU, USRP B210, Pegatron O-RU, MTK UE/ELT, and NVIDIA Aerial L1.
- Executed end-to-end attack detection and mitigation on an OAI UE/gNB testbed, integrating the O1 adapter, rApp, ROMF, WebUI, and K9s controller workflow while preserving evidence-bounded results.

### PT. Pharos Indonesia

AI, Mechatronics & Full-Stack Software Intern

South Jakarta, Indonesia

Jan 2026 – Jun 2026

- Developed an Autoreject Blister inspection application using PyQt, OpenCV, YOLO, and Arduino integration, with interactive ROI/lane editing, CLAHE preprocessing, per-lane counters, product master data, and audit trail.
- Trained and compared YOLO models for pharmaceutical blister quality control, then integrated camera, proximity sensor, servo/rejector, and hardware-debug workflows for local validation.
- Built QR-BPOM production-scan features across a Next.js frontend and Go backend, including WebSocket device events, temporary Master Box flow, duplicate/unreadable QR handling, scan ID reset, and monitoring fixes.

### Fakultas Teknik Universitas Indonesia (FTUI)

Laboratory & Teaching Assistant

Depok, Indonesia

Feb 2025 – Jan 2026

- Laboratory Assistant for Physics Lab:** Facilitated experiments twice a week and graded technical reports for mechanical and electrical physics modules.
- Teaching Assistant for Computational Thinking & Electrical Circuits:** Mentored and graded students in algorithmic logic and Python programming, and supported instruction in circuit analysis.

### Ikatan Mahasiswa Elektro FTUI

Vice Head of Academics and Professions

Depok, Indonesia

Jan 2024 – Dec 2025

- Led strategic academic initiatives and professional development programs for the student body.
- Previously served as Staff of Academics (2024), coordinating peer tutoring and study resources.

## PROJECTS

### O.A.S.I.S. Water Purification System | ESP32, IoT, Power Electronics, RO Filtration

- Designed a portable, off-grid water purification system for post-disaster use, integrating RO filtration, ESP32-based pH/turbidity sensing, adaptive sanitation/drinking-water routing, and offline local monitoring.
- Developed the system architecture and feasibility analysis for a 12V VRLA power train, buck/boost conversion, pumps, solenoid valves, an IP54 enclosure, and a Rp 2.0M bill-of-materials constraint.

### WikiScrolls App | Flutter, Go, Neo4j, Docker, LLM

- Co-developed an educational platform transforming Wikipedia articles into "TikTok-style" audiovisual content.
- Implemented frontend using Flutter and integrated LLM summarization with Text-to-Speech engines.

### Akpro IME Web | Astro, TypeScript, Figma

- Designed UI/UX and developed the frontend for a centralized academic hub used by engineering students.
- Streamlined access to modules, schedules, and curriculum info, improving resource accessibility.

### Draw Battle Game | React, Express.js, PostgreSQL, Socket.io

- Built a real-time multiplayer drawing and guessing game featuring live chat and synchronized canvas.
- Managed full-stack integration between React frontend and PostgreSQL backend via Socket.io.

### Custom Linux Shell | C Language, Linux CLI

- Developed a functional CLI shell handling command execution, foreground/background process management, and IPC via pipes.

### Network Infrastructure Design | Cisco Packet Tracer, OSPF, EIGRP

- Campus Network: Designed VLANs, subnetting, and OSPF routing for university-scale connectivity.
- WAN Architecture: Simulated Frame Relay and PPP topologies connecting HQ to branch offices.

### **AES-128 Hardware Accelerator** | *VHDL, ModelSim, Quartus Prime*

- Designed a high-performance hardware accelerator for AES-128 encryption.
- Authored VHDL code for the core encryption process and validated integrity via testbenches.

### **Smart Drop-Box** | *C++ (ESP32), FreeRTOS, Blynk IoT*

- Architected a secure package delivery system with OTP authentication and cloud monitoring.
- Orchestrated system concurrency using FreeRTOS and developed the mobile interface via Blynk.

### **Noir - Noise & Air Quality Monitor** | *AVR Assembly, Proteus*

- Programmed sensor data processing in Assembly to control audio/visual alerts for environmental monitoring.

### **Parking Counter System** | *Digital Logic, Proteus*

- Designed a logic circuit using counters and flip-flops to track parking availability in real-time.

## TECHNICAL SKILLS

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**Languages:** C/C++, C#, Java, Python, JavaScript, TypeScript, Go, Dart, VHDL, Assembly, LaTeX, Markdown

**Web Development:** React, Next.js, Astro, Express.js, FastAPI, Django, Flutter, HTML/CSS (Tailwind)

**Database:** PostgreSQL, MySQL, MongoDB, Redis, Neo4j

**AI/ML & Vision:** TensorFlow, PyTorch, OpenCV, Ultralytics YOLO

**Networking & Cloud:** Docker, Kubernetes, O-RAN, OpenAirInterface, FlexRIC, NVIDIA Aerial, Cisco Packet Tracer

**Hardware & IoT:** ESP32, Arduino, FreeRTOS, Proteus, ModelSim

**Tools/Design:** Git, Figma, Canva, Adobe, Trello, Jira

## CERTIFICATIONS & AWARDS

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**English Proficiency:** TOEFL ITP: 627 — EPT (UI): 643 (Index A)

**Competitions:** Technoskill 1.0 Web Dev (PIPTEK IME FTUI), Exertion Competitive Programming (EXERCISE FTUI)

**Science Olympiads:** Winner of Amaze with Science, National Science Olympiad (OSN) Math Participant